Aditya Kandukoori

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EDUCATION

Poolesville High School — SMCS Magnet Program **Poolesville, MD**

Expected High School Diploma | Unweighted GPA: 3.93 *August 2024 – June 2028*

Completed Coursework: Magnet Analysis (post-BC calculus), AP Calculus (score: 5), AP Computer Science A, AP Biology, AP U.S. Government

Junior Year Coursework (2026–27): Linear Algebra, Discrete Mathematics, Applied Statistics, Introduction to Artificial Intelligence, AP Physics C (Mechanics & E&M), AP English Language, AP World History

RESEARCH EXPERIENCE

Algoverse AI Research Program **Remote**

Student Researcher — Clinical LLM Safety *May 2026 – Present*

- Authored three research proposals on clinical LLM safety; lead proposal selected by the team over competing submissions for presentation to the faculty PI.
- Designed a novel agentic evaluation protocol testing whether clinical LLM agents retrieve missing patient information before recommending contraindicated medications, extending the RxSafeBench benchmark (2,443 scenarios) from static multiple-choice into tool-augmented decision-making.
- Architected a ReAct-style agent harness with patient-history and drug-database tools, scoring tool-invocation and recommendation safety as independent variables to isolate a failure mode no prior clinical benchmark has measured.
- Conducted the full literature review across five benchmarks (RxSafeBench, AgentClinic, Agent-SafetyBench, npj Digital Medicine, Mind the GAP) establishing the research gap; defined evaluation metrics, statistical tests, and ablations. Target venue: NeurIPS / ML4H workshop.

MoCo Innovation **Montgomery County, MD**

STEM Instructor *2024 – Present*

- Taught hands-on STEM classes to elementary students at Clarksburg Elementary School through a nonprofit serving underrepresented communities.
- Designed the lesson curriculum and instructional materials used in classroom delivery.

PROJECTS

AUMP: Machine Learning Early-Warning System for Crashing Bradycardia

Sole Developer • Kaggle Gemma 4 Hackathon Submission • github.com/akandukoori123-bit/AUMP

Designed, built, and validated an end-to-end ML system predicting bradycardic cardiac events earlier than standard threshold monitors, achieving a median +3-step lead time (Wilcoxon paired test, $p = 1.4 \times 10^{-14}$) at 100% sensitivity and 70.7% specificity across 200 evaluation scenarios. Engineered temporal features (trend, variability, acceleration) from heart-rate time series; trained and benchmarked logistic regression and random forest models with ROC-AUC evaluation. Integrated Gemma 4 via native function calling to build an agentic clinical alert layer with three registered tools, producing structured severity assessments, contraindication detection (e.g., atropine with narrow-angle glaucoma), and multilingual output.

BUMP Labs — Wearable Closed-Loop Cardiac Response Device

Co-Founder • bump-labs.com • Patent Pending (USPTO provisional #63/686,176)

Co-developed the original concept for a wearable atropine pump treating crashing bradycardia and contributed to the device codebase. Currently building an early-stage venture around the system, integrating the AUMP machine learning prediction layer with the BUMP hardware to create a closed-loop predict-and-treat platform.

Comparative Analysis of Digital Tools and Traditional Teaching Methods

Published Research Paper • arXiv:2408.06689 • Second Author

Co-authored a controlled study of 30 elementary students evaluating math skill gains from interactive digital platforms against worksheet-based instruction. Contributed to study design, assessment administration, and data analysis. Digital-tool students improved 24.2% on average versus 8.3% for the control group, demonstrating measurable gains from technology-assisted instruction.

Falconwing — EMG-Controlled Robotic Prosthetic Arm

Built a 3D-printed, Arduino-based prosthetic arm using MyoWare EMG sensors to translate real-time muscle signals into a controlled grasping mechanism.

ACTIVITIES & SKILLS

- **Activities:** Taekwondo — Black Belt, Han Mi Martial Arts (Damascus, MD); Model United Nations — Delegate; 200+ volunteer service hours
- **Programming Languages:** Python, Java
- **ML & Data Science:** Scikit-learn, PyTorch, Pandas, NumPy, Hugging Face Transformers, Matplotlib
- **AI & Agentic Systems:** LLM function calling & tool use, ReAct agent architectures, prompt engineering, LLM evaluation pipeline design
- **Statistical Methods:** Time-series feature engineering, hypothesis testing (Wilcoxon, McNemar's, chi-square), ROC/AUPRC evaluation, experimental design
- **Tools & Environments:** Git/GitHub, Jupyter Notebook, Kaggle, Google Colab, Arduino